

MGT 453 Supply Chain Management

Summer 2026

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OFFICE HOURS: Monday 5-6 PM (Other times by appointment)

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DESCRIPTION

Supply Chain Management involves the flows of materials and information among all of the firms that contribute value to a product, from the source of raw materials to end customers. We will integrate issues from marketing (channels of distribution), logistics, and operations management to develop a broad understanding of a supply chain. Naturally, this course draws upon concepts originally introduced in your pre-term courses on operations management, statistics, marketing, and strategy. While many operational processes that we will consider in this course will appear familiar, most of you will find yourselves viewing them from a different perspective especially through different conceptual frameworks for analyzing supply chain issues.

Data analytics and digital transformation of supply chains became even more important due to digitization movement of the supply chains. Digital transformation also enabled information visibility and transparency across the supply chains. Information visibility helps to speed up the movement of materials including inventory, which saves capital investment and generates free cash flows. This course also covers how data analytics can help supply chains to be more flexible and resilient, and how to evaluate and mitigate risk exposures through analytics.

OBJECTIVES

At the close of this course, you will be able to:

- Understand how supply chains work and describes the major challenges in managing an efficient supply chain through data analytics
- Grasp various strategic and tactical supply chain issues such as distribution and fulfillment strategy, demand forecasting, information sharing strategy, outsourcing, procurement (including e-markets), virtual integration, and risk management.
- Explore supply chain business opportunities from the digitization of supply chains and the emergence of platforms.

MATERIALS

Required

- Supply Chain Management: Strategy, Planning, and Operation 7th Edition, by Sunil Chopra.

SCHEDULE

Mondays 6:30PM – 9:30PM, TBA

COURSE SCHEDULE (Temporary)

Week	Topic(s)	Reading
1	[Strategic Framework for Supply Chain Management] • Course introduction • Strategic Fit	Chapters 1, 2
2	[Supply Chain Metrics and Distribution Network Choice] • Link btw operations and finance • Metrics • Supply chain decision framework • Delivery network design comparison	Chapters 3, 4 HW1: Walmart case (ch3) Case1: Blue Nile case (ch4)
3~4	[Demand Forecasting] • Basic concept of Forecasting • Forecasting methods and error management	Chapter 7 Case 2: SPC (ch7)
5	[Aligning Forecasting with Operations] • Model for aggregate planning • Different Scenarios for SOP	Chapters 8, 9 HW2: Nintendo Game Girl (ch9)
6	[Supply Chain Coordination and Cycle Inventory] • Coordination in a supply chain • “Economies of scale” order quantity	Chapters 10, 11 Quiz
7, 8	[Managing Demand Uncertainty] • Safety stock • Service level and fill rate • Optimal product availability	Chapters 12, 13 Case 3: Penang Electronics (ch12)
9	[Supply Chain Network Design] • Supply network structure • Mathematical model for network design	Chapters 5, 6
10	Guest Speaker, Course Review and Final Exam (take-home)	

GRADING

Component	Percentage
Class Participation	10%
Quiz	10%
Case Write-ups	30%
Homework Assignments	20%
Final Exam	30%
Total	100%

Homework Assignments: C - Collaboration with classmates is allowed. However, each student must submit their own version.

There will be **two** homework assignments throughout the quarter. Each assignment is due ***at the beginning*** of class on its due date since we may cover material on the homework during the class session. Although you may work with your study group on homework assignments, you must individually write-up your own solution to be submitted.

Reading Assignments: No submission of report is required for reading assignments.

For each class session, I will post a set of guiding questions, discussion points, and goals for the lecture on Canvas. You are expected to prepare answers to these questions and be able to provide substantiated arguments on discussion points. It is required that you read the assigned readings before coming to class. This will help give you perspective on the topics to be covered.

Case Write-ups: G – Students may work together in groups and turn in one report per each case for the entire group.

Each group will submit **three** case reports throughout the course. The report can be written in the style of an executive summary (1~2 pages) plus all the detailed calculations or supporting materials. Be to the point and try to imagine yourself to present the result in front of executives. Each group will submit the report with an appendix via Canvas. Each group also needs to submit the report (hard copy) in class when the case is due.

Cases may not be discussed with anyone outside your study group. You may not refer to any reports or spreadsheets from past classes or Internet. It is extremely important and part of the honor code that each member of a group contributes to each case analysis of the group. We may survey peer evaluations around the end of the course to ensure the equal participation of all the team members. Case write-up grades can be adjusted according to the result of the peer evaluations.

All the detailed calculations, tables, and figures can be added as an appendix. The font size can be 11, and 1.5 spacing is recommended. **One or two teams will present their analysis of each case on the day when it is due.**

Quiz: I – Independent, individual work only. No collaboration or consultation allowed.

We will have **one quiz in week 6.**

Exams: I – Independent, individual work only. No collaboration or consultation allowed.

We have the take-home final exam.

Class Participation

Your class participation grade is based upon **your contribution to the discussions** and **attendance**. You are expected to be an active participant throughout the entire class and to contribute to the quality of the discussion. Please note that the frequency of your interventions in class is not a key criterion for effective class contribution. Some criteria used to evaluate class contribution are as follows:

1. Is the participant a good listener?
2. Are the points that are made relevant to the discussion? Are they linked to the comments of others?
3. Do the comments show evidence of analysis of the case?

4. Do comments clarify or build upon the important aspects of earlier comments and lead to a clearer understanding of the case?

ACADEMIC INTEGRITY

Integrity of scholarship is essential for an academic community. As members of the Rady School, we pledge ourselves to uphold the highest ethical standards. The University expects that both faculty and students will honor this principle and in so doing protect the validity of University intellectual work. For students, this means that all academic work will be done by the individual to whom it is assigned, without unauthorized aid of any kind.

The complete UCSD Policy on Integrity of Scholarship can be viewed at:
<http://senate.ucsd.edu/Operating-Procedures/Senate-Manual/Appendices/2>

STUDENTS WITH DISABILITIES

A student who has a disability or special need and requires an accommodation in order to have equal access to the classroom must register with the Office for Students with Disabilities (OSD). The OSD will determine what accommodations may be made and provide the necessary documentation to present to the faculty member.

The student must present the OSD letter of certification and OSD accommodation recommendation to the appropriate faculty member in order to initiate the request for accommodation in classes, examinations, or other academic program activities. **No accommodations can be implemented retroactively.**

Please visit the [OSD website](#) for further information or contact the Office for Students with Disabilities at (858) 534-4382 or osd@ucsd.edu.